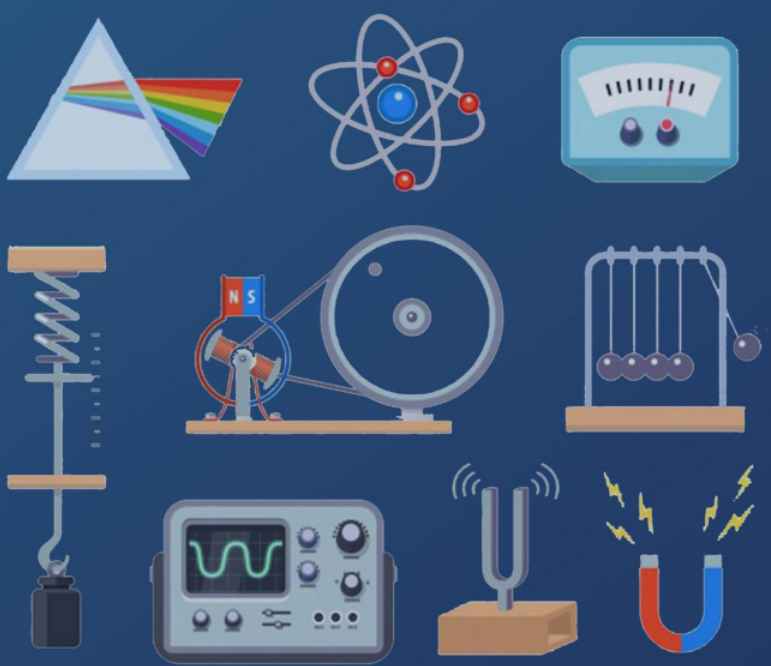
PHYSICS

Wave on String



SHEET

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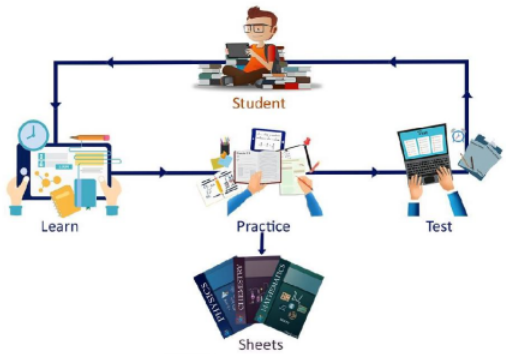
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Note

Detailed solutions are available on the eSaral App.

Module Description



For mastering the concepts only learning is not sufficient. We have to practice and apply those concepts in problem solving. This sheet does just that. It contains a collection of problems segregated in the following exercises to help you master the concepts in a systematic and organized way.

"Practice makes a man perfect"

1. Concept builder – 1 & 1A

As soon as you have finished learning the concept do the problems from these exercises first. These exercises contains easy level questions to help you build your concepts.

1 ⟶ Contains Single Correct Type questions

1A ⟶ Contains pattern based questions incorporating the latest JEE Advanced based patterns like more than one correct, matching list, match the column, etc.

2. Brain Booster – 2 & 2A

Now that you have built your concepts it's time to master them by solving Brain Boosting problems. Don't hurry through these problems. Take time to solve & learn from them. These exercises contains Medium & Tough level problems.

Do questions from 2 & 2A after attempting 1 & 1A

2 ⟶ Single Correct Type questions.

2A ⟶ Pattern Based questions.

3. Simulator – JM & JA

Contains questions from previous year JEE Mains & JEE Advanced questions in exercise JM & exercise JA respectively. Get the real taste & feel of the type of questions being asked in JEE. It's a great tool for simulating your mind with JEE problems.

These exercises are not included in the module but are provided separately.

JM ⟶ JEE Mains previous years topic wise questions.

JA ⟶ JEE Advance previous years topic wise questions.

Home Work Index

Problem solving is an integral part of learning.

Find questions to solve after each video in the homework Index. Make sure that you attempt all the problems (in Ex 1 to 2A) after learning a topic from the videos. For example if you have finished topic 8, first attempt all the problems listed in the index corresponding to topic 8 before proceeding to the video of topic 9.

Sr.No.	Topic Name	Ex-1	Ex-1A	Ex-2	Ex-2A
1	Wave introduction				
2	Wave Function	1,2	1,2		10-12
3	Sine wave travelling on a string	3-7	3-5,11	1-4	
4	Velocity of wave on string	8,9	8	5,6	19,20
5	Power transmission of travelling wave			7,8	
6	Interference of waves			9,10	
7	Transmission of waves			11-13	13-15
8	Standing waves	10			
9	General equation of standing wave			14	1
10	Normal modes on strings	11-13	9	15-20	2-6
11	Energy in standing waves		6,7		7,16-18
12	After full Chapter		10		8,9

Exercise - 1 — Concept Builder

SINGLE CORRECT TYPE

*Each question has FOUR options for correct answer. **ONLY ONE** of these four options is correct option. For each question, choose the correct option to answer the question.*

Wave Function

Q1. At x=0 particle oscillate by law $y=\frac{3}{2t^2+1}$. If wave is propagating along –ve x axis with velocity 2m/s. Find equation of wave

(A) $y=\frac{3}{2\left(t-\frac{x}{2}\right)^2+1}$

(B) $y=\frac{3}{2\left(t+\frac{x}{2}\right)^2+1}$

(C) $y=\frac{3}{2\left(t-\frac{z}{2}\right)^2+1}$

(D) $y=\frac{3}{2\left(t+\frac{z}{2}\right)^2+1}$

Q2. The shape of a wave propagating in the positive x or negative x- direction is given $y=\frac{1}{\sqrt{1+x^2}}$ at t =0 and $y=\frac{1}{\sqrt{2-2x+x^2}}$ at t = 1s where x and y are in meters. The shape the wave disturbance does not change during propagation. Find the velocity of the wave.

(A) 1 m/s in positive x direction

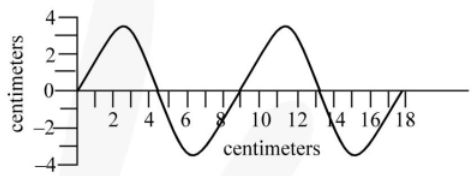
(B) 1 m/s in negative x direction

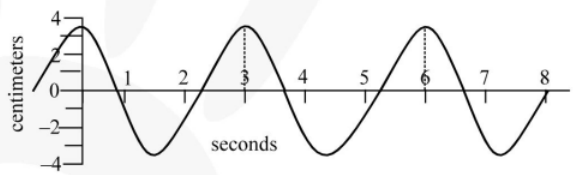
(C) $\frac{1}{2}$ m/s in positive x direction

(D) $\frac{1}{2}$ m/s in negative x direction

Sine Wave Travelling on a String

Q3. A transverse wave is travelling along a horizontal string. The first picture shows the shape of the string at an instant of time. This picture is superimposed on a coordinate system to help you make any necessary measurements. The second picture is a graph of the vertical displacement of *one* point along the string as a function of time. How far does this wave travel along the string in one second ?

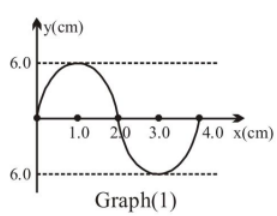




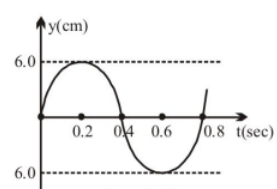
(A) 0.3 cm (B) 3.0 cm

(C) 9.0 cm (D) 27 cm

Q4. Two graphs of the same harmonic wave are shown below. The graph (1) on the left shows the displacement of wave y, as a function of position x for a given instant of time. The graph (2) on the right shows the displacement of wave as a function ot time 't' for a given position. The speed of the wave is :-



Graph(1)

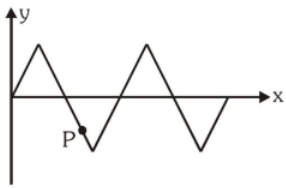


Graph(2)

(A) 5.0 cms^{-1} (B) 0.5 cms^{-1}

(C) 0.4 cms^{-1} (D) 4.0 cms^{-1}

Q5. A triangular transvers wave is propagating in the positive X-direction Velocity of P at this instant will be-



(A) Vertically upward
(B) Vertically downward
(C) At Rest
(D) Cannot be determined

Q6. The equation of a progressive wave for a wire is :

$$Y = 4\sin\left[\frac{\pi}{2}\left(8t - \frac{x}{8}\right)\right]$$

If x and y are measured in cm then velocity of wave is :
(A) 64 cm/s along - x direction
(B) 32 cm/s along - x direction
(C) 32 cm/s along + x direction
(D) 64 cm/s along + x direction

Q7. A plane wave is described by the equation

$$y = 3\cos\left(\frac{x}{4} - 10t - \frac{\pi}{2}\right).$$

The maximum velocity of the particles of the medium due to this wave is

(A) 30 (B) $\frac{3\pi}{2}$ (C) 3/4 (D) 40

Velocity of Wave on String

Q8. Transverse wave of some frequency are generated in two steel wires A and B. The diameter of A is twice of B and the tension in A is half that in B The ratio of velocities of waves in A and B is :-

(A) $1:\sqrt{2}$ (B) 1 : 2
(C) $3:2\sqrt{2}$ (D) $1:2\sqrt{2}$

Q9. The equation of a wave on a string of linear mass density 0.04 kg m^{-1} is given by

$$y = 0.02(m)\sin\left[2\pi\left(\frac{t}{0.04(s)} - \frac{x}{0.50(m)}\right)\right].$$

The tension in the string is :
(A) 6.25 N (B) 4.0 N
(C) 12.5 N (D) 0.5 N

Standing Waves

Q10. A wave is represented by the equation y = 10 sin 2π(100t – 0.02x) + 10 sin 2π(100t + 0.02x). The maximum amplitude and loop length are respectively
(A) 20 units and 30 units
(B) 20 units and 25 units
(C) 30 units and 20 units
(D) 25 units and 20 units

Normal Modes on Strings

Q11. The speed of transverse waves in a stretched string is 700 cm/s. If the string is 2 m long, the frequency with which it resonantes in fundamental mode is :
(A) (7/2) Hz (B) (7/4) Hz
(C) (14) Hz (D) (2/7) Hz

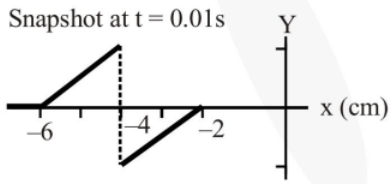
Q12. A string is rigidly tied at two ends and its equation of vibration is given by y = cos2πt sin2πx. Then minimum length of string is :-
(A) 1m (B) 1/2m
(C) 5m (D) 2π m

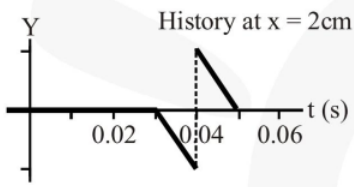
Q13. If the tension in a sonometer wire is increased by a factor of four. The fundamental frequency of vibration changes by a factor of :
(A) 4 (B) (1/4)
(C) 2 (D) (1/2)

ONE OR MORE THAN ONE CORRECT TYPE

Each question has FOUR options for correct answer(s). ***ONE OR MORE THAN ONE*** *of these four option(s) is (are) correct option(s). For each question, choose the correct option(s) to answer the question.*

Q1. Figure shows a snapshot graph and a history graph for a wave pulse on a stretched string. They describe the same wave from two perspectives.





(A) The wave is travelling in positive x-direction

(B) The wave is travelling in negative x-direction

(C) The speed of the wave is 2 m/s.

(D) The peak is located at x = –6 cm at t = 0.

Q2. $Y(x,t) = 0.05 / [(4x + 2t)^2 + 5]$ represents a moving wave pulse, where x and y are in meters and t is in seconds. Then which statement(s) are **CORRECT**:

(A) Pulse is moving in – x direction

(B) Wave speed is 0.5 m/s

(C) Maximum particle displacement is 1 cm

(D) It is a symmetric pulse

Q3. The mathematical forms for three sinusoidal traveling waves are given by

Wave 1 : $y(x,t) = (2cm) \sin(3x–6t)$

Wave 2 : $y(x,t) = (3cm) \sin(4x–12t)$

Wave 3 : $y(x,t) = (4cm) \sin(5x–11t)$

where x is in meters and t is in seconds.

Of these waves :

(A) Wave 1 has the greatest wave speed and the greatest maximum transverse string speed.

(B) Wave 2 has the greatest wave speed and wave 1 has the greatest maximum transverse string speed.

(C) Wave 3 has the greatest wave speed and the greatest maximum transverse string speed.

(D) Wave 2 has the greatest wave speed and wave 3 has the greatest maximum transverse string speed.

Q4. A transverse wave travelling on a taut string is represented by $y = 0.01 \sin2\pi (10 t – x)$, y and x are in meters and t in seconds. Then:-

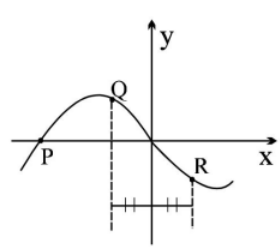
(A) The speed of the wave is 10 m/s.

(B) Closest points on the string which differ in phase by 60° are (1/6) m apart.

(C) Maximum particle velocity is $\pi/4$ m/s.

(D) The phase of a certain point on the string changes by 120° in (1/20) seconds.

Q5. At a certain moment, the photograph of a string on which a harmonic wave is travelling to the right is shown. Then, which of the following is true regarding the velocities of the points P, Q and R on the string.



(A) v_P is upwards
(B) $v_Q = -v_R$
(C) $|v_P| > |v_Q| = |v_R|$
(D) $v_Q = v_R$

Q6. A standing wave of time period T is set up in a string clamped between two rigid supports. At t = 0 antinode is at its maximum displacement 2A.
(A) The energy density of a node is equal to energy density of an antinode for the first time at t = T/4.
(B) The energy density of node and antinode becomes equal after T/2 second.
(C) The displacement of the particle at antinode at $t = \frac{T}{8}$ is $\sqrt{2}A$.
(D) The displacement of the particle at node is zero.

Q7. A clamped string is oscillating in nth harmonic, then :-
(A) Total energy of oscillations will be proportional to n^2 times that of fundamental frequency
(B) Total energy of oscillations will be proportional to $(n-1)^2$ times that of fundamental frequency
(C) Average kinetic energy of the string over a complete oscillations is half of that of the total energy of the string.
(D) None of these

NUMERICAL TYPE

*The answer to each question is a **NUMERICAL VALUE**. For each question, find the correct numerical value (in decimal notation, truncated/ rounded-off to the second decimal place; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30)*

Q8. A wave moves with speed 300 ms^{-1} on a wire which is under a tension of 500 N. Find how much the tension must be changed to increase the speed to 360 ms^{-1}.

Q9. A sonometer wire emits a fundamental note of frequency 150 Hz. Calculate the frequency of the note emitted when the tension is changed in the ratio of 9 : 16 and length in the ratio of 1 : 2.

Q10. A wave has a speed of 300 m/sec and frequency 500 Hz. The phase difference between two adjacent points is π/3 radian. What will be the path difference between them in cm?

SUBJECTIVE TYPE

Q11. The displacement produced by a simple harmonic wave is $y = \frac{10}{\pi}\sin\left(200\pi t - \frac{\pi x}{17}\right)$. Then find the time period and maximum velocity of the particle.
(x → m, y → cm, t → sec.)

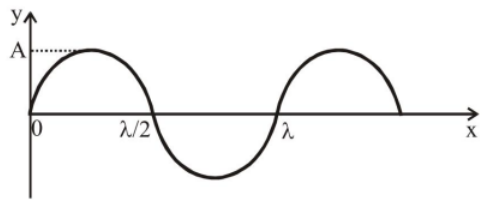
Exercise - 2 *Brain Booster*

SINGLE CORRECT TYPE

*Each question has FOUR options for correct answer. **ONLY ONE** of these four options is correct option. For each question, choose the correct option to answer the question.*

Q1. Here given snap shot of a progressive wave at t = 0 with time period = T. Then the equation of the wave if wave is going in +ve x-direction and if wave is going in –ve x-direction will be respectively.

$\left(\text{Here, } T = \frac{2\pi}{\omega}\right)$



(A) $y = A\sin(kx + \omega t),\ y = A\sin(kx - \omega t)$

(B) $y = A\cos(kx + \omega t),\ y = A\cos(kx - \omega t)$

(C) $y = A\sin(\omega t - kx),\ y = A\sin(\omega t + kx)$

(D) $y = A\sin(kx - \omega t),\ y = A\sin(kx + \omega t)$

Q2. A sinusoidal wave travelling in the positive direction of x on a stretched string has amplitude 2.0 cm, wavelength 1 m and wave velocity 5.0 m/s. At x = 0 and t =0, it is given that displacement y = 0 and $\frac{\partial y}{\partial t} < 0$. Express the wave function correctly in the form y = f(x, t) :-

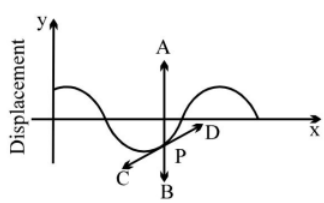
(A) $y = (0.02\text{ m})\sin 2\pi(x - 5t)$

(B) $y = (0.02\text{ cm})\cos 2\pi(x - 5t)$

(C) $y = (0.02\text{ m})\sin 2\pi\left(x - 5t + \frac{1}{4}\right)$

(D) $y = (0.02\text{ cm})\cos 2\pi\left(x - 5t + \frac{1}{4}\right)$

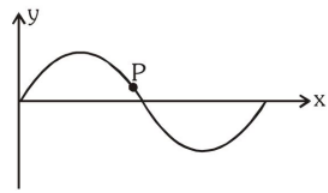
Q3. The figure below shows a snap photograph of a simple harmonic progressive wave, progressing in the negative X-axis, at a given instant. The direction of the velocity of the particle at the stage P on the figure is best represented by the arrow.



(A) $\vec{PA}$ (B) $\vec{PB}$

(C) $\vec{PC}$ (D) $\vec{PD}$

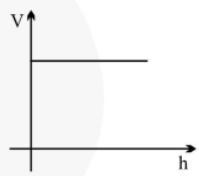
Q4. A transverse sinusoidal wave moves along a string in the positive x-direction at a speed of 10 cm/s. The wavelength of the wave is 0.5 m and its amplitude is 10 cm. At a particular time t, the snap shot of the wave is shown in figure. The velocity of P when its displacement is 5 cm is :

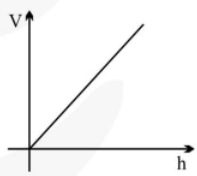


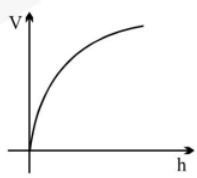
(A) $\frac{\sqrt{3}\pi}{50}\hat{i}$ m/s (B) $-\frac{\sqrt{3}\pi}{50}\hat{j}$ m/s

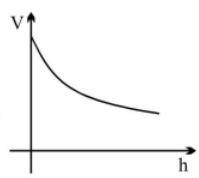
(C) $\frac{\sqrt{3}\pi}{50}\hat{j}$ m/s (D) $-\frac{\sqrt{3}\pi}{50}\hat{i}$ m/s

Q5. A uniform rope having some mass hanges vertically from a rigid support. A transverse wave pulse is produced at the lower end. The speed (v) of the wave pulse varies with height (h) from the lower end as:

(A)


(B)


(C)


(D)


Q6. The extension in a string, obeying Hookes' law, is x. The speed of the wave in the stretched string is v. If the extension in the string is increased to 1.5 x, the speed of the wave in the string will be

(A) $1.22v$ (B) $0.61v$

(C) $1.50\ v$ (D) $0.75v$

Q7. The average power transmitted by a progressive wave in term of standard notations is :-

(A) $4\pi^2 A^2 \mu^{1/2} T^{-1/2} f^2$

(B) $2\pi^2 A^2 \mu^{1/2} T^{1/2} f^2$

(C) $2\pi^2 A^2 \mu\ T^{1/2} f^2$

(D) None

Q8. The prong of a electrically operated tuning fork is connected to a long string of $\mu = 1$ kg/m and tension 25 N. The maximum velocity of the prong is 1 cm/s, then the average power needed to drive the prong is:



(A) 5×10^{-4} W (B) 2.5×10^{-4} W

(C) 1×10^{-4} W (D) 10^{-3} W

Q9. Five waveforms moving with equal speeds on the x-axis

$y_1 = 8 \sin(\omega t + kx)$;

$y_2 = 6 \sin(\omega t + \frac{\pi}{2} + kx)$;

$y_3 = 4 \sin(\omega t + \pi + kx)$;

$y_4 = 2 \sin(\omega t + \frac{3\pi}{2} + kx)$;

$y_5 = 4\sqrt{2} \sin(\omega t - kx + \frac{\pi}{4})$

are superimposed on each other. The resulting wave is :

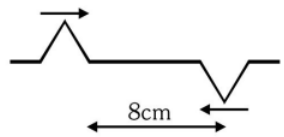
(A) $8\sqrt{2} \cos kx \sin(\omega t + \frac{\pi}{4})$

(B) $8\sqrt{2} \sin(\omega t - kx + \frac{\pi}{4})$

(C) $8\sqrt{2} \sin kx \cos(\omega t + \frac{\pi}{4})$

(D) $8 \sin(\omega t + kx)$

Q10. Two pulses in a stretched string, whose centres are initially 8 cm apart, are moving towards each other as shown in the figure. The speed of each pulse is 2 cm/s. After 2 s the total energy of the pulses will be :-



(A) Zero

(B) Purely kinetic

(C) Purely potential

(D) Partialy kinetic and partialy potential

Q11. A wave of frequency 100 Hz travels along a string towards its fixed end when this wave travels back, after reflection a node is formed at a distance of 10 cm from the fixed end. The speed of the wave (incident and reflected) is :

(A) 5 m/s (B) 10 m/s

(C) 20 m/s (D) 40 m/s

Q12. The equation of a plane progressive wave is $y = 0.02 \sin 8\pi \left[t - \frac{x}{20}\right]$. When it is reflected at a rarer medium (medium with higher velocity) at $x = 0$, its amplitude becomes 75% of its previous value. The equation of the reflected wave is :-

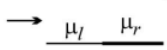
(A) $y = 0.02 \sin 8\pi \left[t - \frac{x}{20}\right]$

(B) $y = 0.02 \sin 8\pi \left[t + \frac{x}{20}\right]$

(C) $y = +0.015 \sin 8\pi \left[t + \frac{x}{20}\right]$

(D) $y = -0.015 \sin 8\pi \left[t + \frac{x}{20}\right]$

Q13. A string consists of two parts attached at x = 0. The right part of the string (x > 0) has mass μ_r per unit length and the left part of the string (x < 0) has mass μ_l per unit length. The string tension is T. If a wave of unit amplitude travels along the left part of the string, as shown in the figure, what is the amplitude of the wave that is transmitted to the right part of the string ?



(A) 1 (B) $\frac{2}{1+\sqrt{\mu_l/\mu_r}}$

(C) $\frac{2\sqrt{\mu_l/\mu_r}}{1+\sqrt{\mu_l/\mu_r}}$ (D) $\frac{\sqrt{\mu_l/\mu_r}-1}{\sqrt{\mu_l/\mu_r}+1}$

Q14. The shape of a string on which standing waves are produced at t = 0 is shown here. The suitable equation of standing wave can be :



(A) y = A sin kx cos ωt

(B) y = A sin kx sin ωt

(C) y = A cos kx sin ωt

(D) y = A cos kx cos ωt

Q15. A steel wire of length 1m, mass 0.1 kg and uniform cross sectional area 10^{-6} m^2 is rigidly fixed at both ends. The temp difference of wire is 20°C. If the transverse waves are set up by plucking the string in the middle, calcutate freq. of fundamental mode of vibration.

(Y = $2x10^{11}$ N/m^2, α = $1.21x10^{-5}$ /°C)

(A) 11 Hz (B) 20 Hz

(C) 22 Hz (D) 15Hz

Q16. A vibrator makes 150 cm of a string to vibrate in 6 loops in the longitudinal arrangement when it is stretched by 150 N. The entire length of the string is then weighed and is found to weigh 400 mg. Then:-

(A) Frequency of the vibrator is 3 kHz.

(B) Frequency of the vibrator is 1.5 kHz.

(C) Distance between two nodes is 25 cm.

(D) Distance between two nodes is 33 cm.

Q17. A sonometer wire resonates with a given tuning fork forming standing waves with five antinodes between the two bridges when a mass of 9kg is suspended from the wire. When this mass is replaced by mass M, the wire resonates with the same tuning fork forming three antinodes for the same position of the bridges. Then find the value of square root of M.

(A) 5 (B) 10

(C) 25 (D) None

Q18. A string of length 3L is fixed at both ends. It resonates with a tuning fork in third harmonic with amplitude at antinode equal to A_0. At time t = 0, a string element at position of antinode is at half its positive amplitude and moving towards mean position. Displacement of a string element at L/2 is given by :-

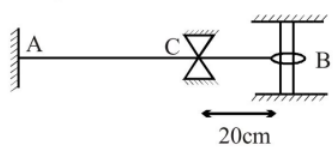
(A) $\frac{A_0}{2}\sin\left(\omega t+\frac{11\pi}{6}\right)$

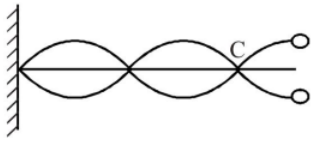
(B) $\frac{\sqrt{3}A_0}{2}\sin\left(\omega t+\frac{5\pi}{6}\right)$

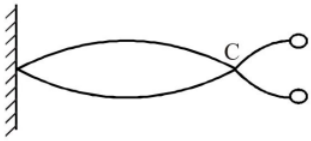
(C) $A_0\sin\left(\omega t+\frac{5\pi}{6}\right)$

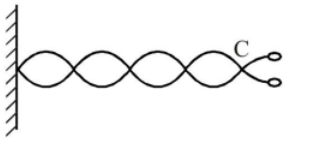
(D) $\frac{A_0}{2}\sin\left(\omega t+\frac{5\pi}{6}\right)$

Q19. A 1m long wire having tension T is fixed at A and free at B. The point C, 20 cm from B is constrained to be stationary. What is shape of string for fundamental mode ?



(A) 

(B) 

(C) 

(D) 

Q20. Two vibrating strings of the same material but of lengths L and 2L have radii 2r and r respectively. They are stretched under the same tension. Both the strings vibrate in their fundamental modes, the one of length L with frequency f_1 and the other with frequency f_2. The ratio $\frac{f_1}{f_2}$ is given by :-

(A) 2 (B) 4
(C) 8 (D) 1

Exercise - 2A *Brain Booster*

ONE OR MORE THAN ONE CORRECT TYPE

Q1. The vibration of a string fixed at both ends are described by $Y= 2 \sin(\pi x) \sin(100\pi t)$ [where Y is in mm, x is in cm and t in second] then

(A) Maximum displacement of the particle at x = 1/6 cm would be 1 mm.

(B) Velocity of the particle at x = 1/6 cm at time t = 1/600 sec will be 157 $\sqrt{3}$ mm/s

(C) If the length of the string be 10 cm, number of loop in it would be 5

(D) None of these

Q2. The tension in a stretched string fixed at both ends is changed by 2%, the fundamental frequency is found to get changed by 15 Hz. Select the correct statement(s)

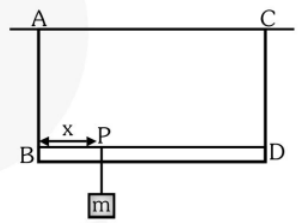
(A) Wavelength of the string of fundamental frequency does not change.

(B) Velocity of propagation of wave changes by 2%

(C) Velocity of propagation of wave changes by 1%

(D) Original frequency is 1500 Hz.

Q3. A massless rod BD is suspended by two identical massless strings AB and CD of equal lengths. A block of mass m is suspended at point P such that BP is equal to x, if the fundamental frequency of the left wire is twice the fundamental frequency of right wire, then the value of x is :-



(A) $\frac{\ell}{5}$ (B) $\frac{\ell}{4}$
(C) $\frac{4\ell}{5}$ (D) $\frac{3\ell}{4}$

Q4. Figure, shows a stationary wave between two fixed points P and Q.

Which point (s) of 1, 2 and 3 are in phase with the point X ?



(A) 1, 2 and 3 (B) 1 and 2 only
(C) 2 and 3 only (D) 3 only

Q5. An string has resonant frequencies given by 1001 Hz and 2639 Hz.

(A) If the string is fixed at one end only, 910 Hz can be a resonance frequency.

(B) If the string is fixed at one end only, 1911 Hz can be a resonance frequency.

(C) If the string is fixed at both the ends, 364 Hz can be one of the resonant frequency.

(D) 1001 Hz is definitely not the fundamental frequency of the string.

Q6. A narrow steel rod of length 'L' is rigidly clamped at its mid-point and transverse standing waves of frequency 'f' are set up in it. The speed of transverse waves in the rod is 'c'. Then :-

(A) The free ends of the rod must be Antinodes

(B) The fundamental frequency 'f' of the rod is c/(L)

(C) The second overtone frequency of the rod is 5c/(2L)

(D) 'f' can be any integral multiple of the fundamental frequency

Q7. The ends of a stretched wire of length L are fixed at x=0 and x = L. In one experiment the displacement of the wire is $y_1 = A\sin\left(\frac{\pi x}{L}\right)\sin\omega t$ and energy is E_1 and in other experiment its displacement is $y_2 = A\sin\left(\frac{2\pi x}{L}\right)\sin 2\omega t$ and energy is E_2.

Then :-

(A) $E_2 = E_1$ (B) $E_2 = 2E_1$

(C) $E_2 = 4E_1$ (D) $E_2 = 16E_1$

Q8. A plane transverse wave is propagating in a direction making an angle of 30° with positive x-axis in the x-y plane. Find phase difference between points (0, 0, 0) and (1, 1, 1). Wavelength of the wave is 1m :-

(A) 2π rad (B) $\left(\sqrt{3}+1\right)\pi$ rad

(C) $\left(\sqrt{2}+1\right)\pi$ rad (D) None

Q9. Consider a hypothetical wave pulse (at time t =0) given by the following (y, x in meter)

$y = 0, x < 0$

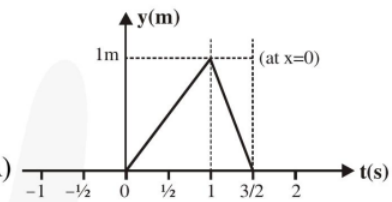
$y = x/2, 2 > x \geq 0$

$y = 3 - x, 3 \geq x \geq 2$

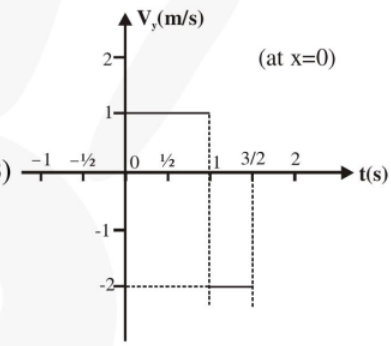
$y = 0, x > 3$

The pulse travels leftwards (negative x direction) at speed v = 2 m/s. Which of the following plots are correct? [V_y is the velocity of particle]

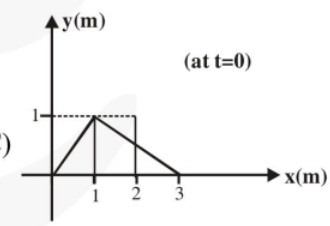
(A)



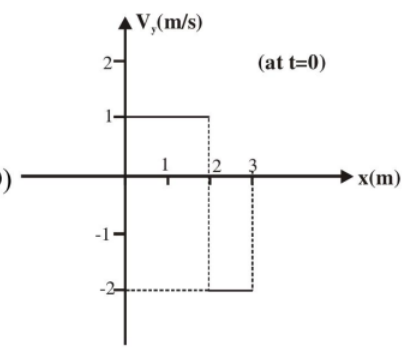
(B)



(C)



(D)

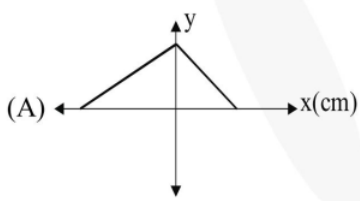


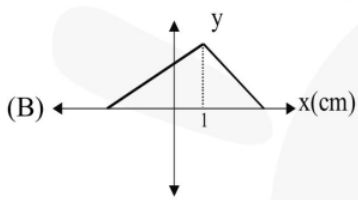
PARAGRAPH TYPE

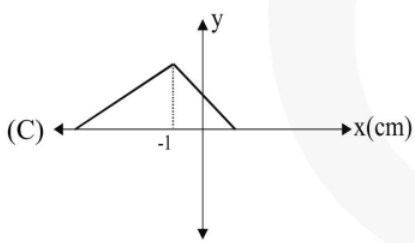
Paragraph for Question Nos. 10 to 12

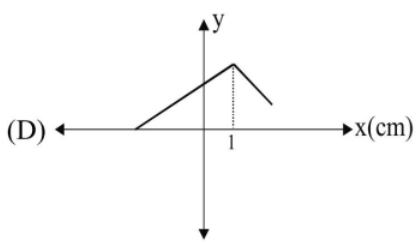
At a time t =0 a pulse moving in the +ve x-direction with velocity v in the shape given by the function between with $y=\begin{cases}\frac{x}{4}+1 & -4<x<0\\ -x+1 & 0<x<1\\ 0 & otherwise\end{cases}$, where x and y are in cm.

Q10. If velocity is 10 m/s, which of the graphs shows the shape and position of the string at a time t = 1 millisec.

(A)


(B)


(C)


(D)


Q11. The wave equation that explicitly gives the displacement of any point at any time is :

(A) $y=\begin{cases}\frac{1}{4}(x-Vt)+1 & -4<(x-Vt)<0\\ -(x-Vt)+1 & 0<(x-Vt)<1\\ 0 & otherwise\end{cases}$

(B) $y=\begin{cases}\frac{1}{4}(x+Vt)+1 & -4<(x+Vt)<0\\ -(x+Vt)+1 & 0<(x+Vt)<1\\ 0 & otherwise\end{cases}$

(C) $y=\begin{cases}-\frac{1}{4}(x+Vt)+1 & -4<(x+Vt)<0\\ (x+Vt)+1 & 0<(x+Vt)<1\\ 0 & otherwise\end{cases}$

(D) $y=\begin{cases}-\frac{1}{4}(x-Vt)+1 & -4<(x-Vt)<0\\ (x-Vt)+1 & 0<(x-Vt)<1\\ 0 & otherwise\end{cases}$

Q12. Velocity of a point at x = 20 m as a function of time is given as

(A) $v=\begin{cases}\frac{1}{4}\,m/s & 1.999<t<2\ s\\ -1\ m/s & 2<t<2.004\ s\\ 0 & otherwise\end{cases}$

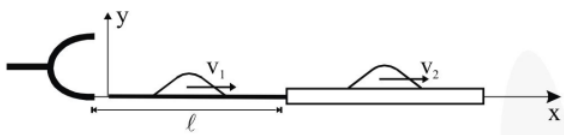
(B) $v=\begin{cases}1m/s & 1.999<t<2\ s\\ -\frac{1}{4}\ m/s & 2<t<2.004\ s\\ 0 & otherwise\end{cases}$

(C) $v=\begin{cases}10m/s & 1.999<t<2\ s\\ -\frac{5}{2}\ m/s & 2<t<2.004\ s\\ 0 & otherwise\end{cases}$

(D) $v=\begin{cases}\frac{5}{2}\,m/s & 1.999<t<2\ s\\ -10\ m/s & 2<t<2.004\ s\\ 0 & otherwise\end{cases}$

Paragraph for Question Nos. 13 to 15

A harmonic oscillator at x = 0, oscillates with a frequency $\frac{\omega}{2\pi}$ and amplitude a. It is generating waves at end of a thin string in which velocity of wave is v_1 and which is connected to another heavier string in which velocity of wave is v_2 as shown, length of first string is ℓ.



Q13. If harmonic oscillator oscillates by an equation y = a sinωt. The equation of incident wave in first string is :-

(A) $y = a\sin\omega\left(t - \frac{x}{v_1}\right)$

(B) $y = a\sin\omega\left(t + \frac{x}{v_1}\right)$

(C) $y = a\sin\left[\omega\left(t - \frac{x}{v_1}\right) + \pi\right]$

(D) $y = a\sin\left[\omega\left(t + \frac{x}{v_1}\right) + \pi\right]$

Q14. Equation of transmitted wave in second string if its amplitude is a_t is :-

(A) $y = a_t\sin\omega\left(t - \frac{x}{v_2}\right)$

(B) $y = a_t\sin\omega\left(t - \frac{\ell}{v_1}\right)$

(C) $y = a_t\sin\omega\left(t - \frac{\ell}{v_1} - \frac{x-\ell}{v_2}\right)$

(D) $y = a_t\sin\omega\left(t - \frac{x}{v_2}\right)$

Q15. Equation of reflected wave, if it is reflecting at the joint and amplitude of reflected wave is a_R.

(A) $y = a_R\sin\omega\left(t - \frac{x}{v_2}\right)$

(B) $y = a_R\sin\left[\omega\left(t - \frac{\ell}{v_1} - \frac{\ell - x}{v_1}\right) + \pi\right]$

(C) $y = a_R\sin\left[\omega\left(t + \frac{x}{v_1}\right) + \pi\right]$

(D) $y = a_R\sin\left[\omega\left(t + \frac{2\ell + x}{v_1}\right) + \pi\right]$

Paragraph for Question Nos. 16 to 18

A wave represented by equation y = 2(mm) sin $[4\pi(\text{sec}^{-1})t - 2\pi(\text{m}^{-1})x]$ is superimposed with another wave y = 2 (mm) sin $[4\pi(\text{sec}^{-1})t + 2\pi(\text{m}^{-1})x + \pi/3]$ on a tight string.

Q16. Phase difference between two particles which are located at $x_1 = 1/7$ and $x_2 = 5/12$ is

(A) 0 (B) $\frac{5\pi}{6}$

(C) π (D) $\frac{5\pi}{3}$

Q17. Which of the following is not a location of antinode?

(A) $\frac{2}{3}$ (B) $\frac{11}{12}$

(C) $\frac{5}{12}$ (D) $\frac{17}{12}$

Q18. The location having maximum potential energy is :

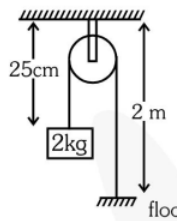
(A) 1/7 (B) 1/6

(C) 5/12 (D) 23/12

NUMERICAL TYPE

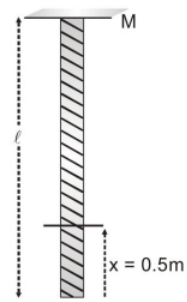
*The answer to each question is a **NUMERICAL VALUE**. For each question, find the correct numerical value (in decimal notation, truncated/ rounded-off to the second decimal place; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30)*

Q19. In the given figure the string has mass 4.5g. Find the time taken by a transverse pulse produced at the floor to reach the pulley. ($g = 10\ ms^{-2}$).



SUBJECTIVE TYPE

Q20. A uniform rope of mass 0.1 kg and length 2.45 m hangs from a ceiliing as shown in diagram.



(a) Find speed of transverse wave in the rope at a point 0.5 m distance from the lower end.

(b) Calculate time taken by a transverse wave to travel the full length of rope.

Answer Key

Ex-1

1	2	3	4	5	6	7	8	9	10
B	A	B	A	A	D	A	D	A	B
11	12	13							
B	B	C							

Ex-1A

1. A,C,D **2.** A,B,C,D **3.** D **4.** A,B **5.** C,D
6. C,D **7.** A,C **8.** 220 N **9.** 100 Hz **10.** 10 cm
11. $\frac{1}{100}$ s, 20m/s

Ex-2

1	2	3	4	5	6	7	8	9	10
D	A	A	C	C	A	B	B	A	B
11	12	13	14	15	16	17	18	19	20
C	C	C	A	A	C	A	C	A	D

Ex-2A

1. A,B **2.** A,C **3.** A **4.** C **5.** B,C,D
6. A,C **7.** C **8.** B **9.** A,B,D **10.** B
11. A **12.** C **13.** A **14.** C **15.** B
16. C **17.** A **18.** B **19.** 0.02 s
20. (a) 2.21 m/sec , (b) 1 sec.